

第一部分·中文版

EPS 54 周报·第 2 周 (2026 年 9 月 13 日 – 9 月 21 日)

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一、本周工作内容

1. 开发环境与工具链

- Python 3.14.5 + 项目虚拟环境，库为 NumPy / pandas / Matplotlib / Jupyter ; VS Code 配好 Python 与 Jupyter 扩展。接入agent辅助讲解代码。版本控制：GitHub 仓库 Gra，链接为changhaiWangDVC12 (dash)。

2. 深度学习基础 (按计划推进)

- 1. 《动手学深度学习》(链接为2.1. 数据操作 — 动手学深度学习 2.0.0 documentation) 第 1–2 章：数据操作 (2.1)、数据预处理 (2.2)、线性代数 (2.3, 含 2.3.7–2.3.10 点积、矩阵-向量积、矩阵乘法、范数)。其中 2.3.7–2.3.10 单独整理成一份中文讲解页，因为这几个运算是风险场公式的语言。
- 2. 2.4 微积分学完：导数、偏导、梯度、链式法则。现在能讲清“风险场是定义在空间上的标量场，负梯度就是风险下降最快的方向，也就是避让方向”。

3. L0 动手练习 1–3 (全部跑通，有代码与图)

- 练习 1 · 自造轨迹数据：两辆车各 100 帧 (10 Hz, 覆盖 9.9 秒)。车 A 在 $y=0$ 车道以 15 m/s 直行；车 B 在相邻车道 $y=3.5$ m 以 12 m/s 直行，初始在 A 前方 60 m。
- 练习 2 · 距离矩阵与 TTC：用广播一次性算出任意两帧之间的 100×100 距离矩阵 D，取对角线得到同帧距离 $d(t)$ ，再除以纵向相对速度 3 m/s 得到 $TTC(t)$ 。全程向量化，没有逐帧 for 循环。
- 练习 3 · BEV 俯视图：x/y 比例 1:1；画出 $y=0$ 与 $y=3.5$ 两条车道中心线、两条轨迹（按速度着色），坐标轴带单位（米）。

4. 一个发现

- 在练习3这个场景里，只按纵向算出来的 TTC 从 20.03 s 降到 10.17 s，看上去“很安全”；但两车始终分处两条车道，同帧最近距离是 30.50 m，而两条车道中心线只相距 3.5 m，在几何上永远不会相撞。也就是说：TTC 只把纵向相对距离与纵向相对速度算进去，横向这一维根本没有进入指标。对这种永远碰不到的场景它给假警报，对真正斜向切入的冲突它又看不见。
- 3.这正好对应任务书里“只关注纵向、不覆盖整个行驶空间”的批评点，也是我打算先用数字做实的第一条基线缺口。

5. 文献

- 开始按方法族读，不按时间流水账。已读 criticality metrics 综述 (arXiv:2108.02403) 的摘要与引言；它的分类表可以直接当我要批评的那一代指标的坐标系。

二、遇到的问题

1. 概念层面 (在补)

- 形状与广播是实操中最容易出错的一环（axis 与 keepdims 搞反不报错但算错）。目前的对策是每次写完都打印形状自检。
- 微积分还没到“能用”的程度：能讲清梯度的意义，但还不能把场类论文里的核心公式改写成自己能跑的函数。计划靠 2.5 自动微分和后续复现来补，目标是后续能把 EDRF 一类方法的核心公式写成 NumPy 函数。

2. 需要导师给方向的（目前卡在这里）

- 数据：driverresearch 数据集的申请走哪个路径？是否需要联系邮箱contact@driverresearch.tech？拿到之前我用自造轨迹练手，并计划先用 highD/inD 把数据格式与代码通路打通。
- 基线：第一版基线先复现 TTC 族，还是直接复现 EDRF / 时空风险场这类场方法？有没有您希望我对标的内部实现或代码？

三、下周计划

1. d2l 2.5 自动微分、2.6 概率，一定读过学过并写出总结。
2. 读三篇综述（2108.02403 / 2202.02215 / 2403.02622 或 2501.11260），每篇一张五行卡片记录内容与心得。
3. 填满 driverresearch 字段表（13 行），弄清 ISO 34502 是干什么的，并按您给的路径提交数据申请。
4. 读一遍 highD 数据格式说明，用它的工具包跑出一张自己的 BEV 图（先不下大文件）。

本周投入时间：____ 小时（计划负荷为每周 8–10 小时）。

Part 2 · English version

EPS 54 Weekly Report — Week 2 (13–21 September 2026)

Student: WANG Yibo (yw2089@hw.ac.uk) | Project: World model informed Spatial-Temporal Risk Field Modeling for Safety Assessment of Autonomous Vehicles | Supervisor: Dr Cheng Wang

1. Work completed this week

1.1 Development environment and toolchain

- Python 3.14.5 with a project virtual environment, and the NumPy / pandas / Matplotlib / Jupyter stack; VS Code configured with the Python and Jupyter extensions. I have an AI agent set up to help explain the code. Version control: repository “Gra” on GitHub (github.com/changhaiWangDVC12, account “dash”).

1.2 Deep learning fundamentals (per plan)

- 1. “Dive into Deep Learning” (link: 2.1. Data Manipulation — Dive into Deep Learning 2.0.0 documentation), Chapters 1–2: data manipulation (2.1), data preprocessing (2.2), linear algebra (2.3, including 2.3.7–2.3.10 — dot product, matrix-vector product, matrix multiplication, norms). Of these, 2.3.7–2.3.10 were organised into a separate Chinese-language study page, because these operations are the language in which risk-field formulas are written.
- 2. 2.4 completed (calculus): derivatives, partial derivatives, gradients and the chain rule. I can now explain clearly that a risk field is a scalar field defined over space, so the negative gradient is the direction in which risk decreases fastest — that is, the avoidance direction.

1.3 Hands-on exercises L0-1 to L0-3 (all working, with code and figures)

- Exercise 1 — synthetic trajectory data: 100 frames per vehicle (10 Hz, covering 9.9 s). Vehicle A travels straight in lane $y = 0$ at 15 m/s; vehicle B travels straight in the adjacent lane $y = 3.5$ m at 12 m/s, initially 60 m ahead of A.
- Exercise 2 — distance matrix and TTC: computed the full 100×100 distance matrix D between any two frames of the two vehicles in a single broadcasting operation, took its diagonal as the same-frame distance $d(t)$, and divided by the longitudinal relative speed (3 m/s) to obtain $TTC(t)$. Fully vectorised, with no per-frame for loop.
- Exercise 3 — bird’s-eye-view (BEV) plot: x/y scaling 1:1; drew both lane centre lines ($y = 0$ and $y = 3.5$ m) and both trajectories (coloured by speed), with the axes labelled in metres.

1.4 A finding

- In the scenario of Exercise 3, the longitudinal-only TTC falls from 20.03 s to 10.17 s, which looks “very safe”; yet the two vehicles remain in two different lanes throughout — their minimum same-frame separation is 30.50 m while the two lane centre lines are only 3.5 m apart, so geometrically they can never collide. In other words: TTC takes into account only the longitudinal relative distance and the longitudinal relative speed; the lateral dimension never enters the metric at all. It raises a false alarm for a scenario that can never be hit, and it is blind to a genuinely oblique cut-in conflict.
- 3. This is exactly the criticism in the project brief (“focuses only on the longitudinal dimension and does not cover the whole driving space”), and it is the first gap in the baseline that I intend to establish with

numbers.

1.5 Literature

- Started reading by method family, not as a chronological log. Have read the abstract and introduction of the criticality-metrics survey (arXiv:2108.02403); its classification table can serve directly as the coordinate system of the generation of metrics I intend to critique.

2. Problems encountered

2.1 Conceptual level (still being filled in)

- Shape and broadcasting are the part of hands-on work in which it is easiest to make a mistake (getting axis and keepdims the wrong way round raises no error but computes the wrong thing). My countermeasure at the moment is to print the shapes as a self-check after everything I write.
- Calculus is not yet at a “usable” level: I can explain the meaning of the gradient, but I cannot yet rewrite the core formulas in field-based papers as functions I can run myself. I plan to close this gap with 2.5 (automatic differentiation) and by reproducing methods later; the goal is to be able to write the core formulas of an EDRF-style method as NumPy functions.

2.2 Where I need guidance from you (currently blocked here)

- Data: which route should the application for the driverresearch dataset take? Do I need to contact the email address contact@driverresearch.tech? Until then I am practising on synthetic trajectories, and I plan to use highD/inD first to get the data format and the code path working.
- Baseline: for the first version, should I reproduce the TTC family, or reproduce a field-based method such as EDRF / spatial-temporal risk fields directly? Is there an internal implementation or code base you would like me to benchmark against?

3. Plan for next week

1. d2l 2.5 (automatic differentiation) and 2.6 (probability): read and worked through, with a written summary.
2. Read three surveys (2108.02403 / 2202.02215 / 2403.02622 or 2501.11260), one five-line card per paper recording the content and my takeaways.
3. Complete the driverresearch field table (13 fields), clarify what ISO 34502 is used for, and submit the data application via the route you advise.
4. Read the highD data format specification and run its toolkit to produce one BEV plot of my own (without downloading the large files first).

Time invested this week: ____ hours (planned load: 8–10 hours per week).